



Team 4

PRODUCTO 1

Conmutación y Enrutamiento

Implementación de Red Empresarial

Presentado por:

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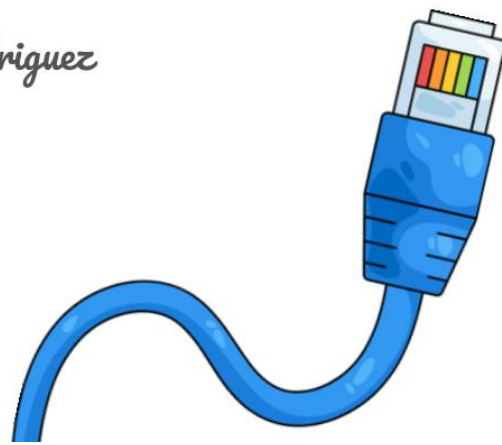
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PROYECTO DE IMPLEMENTACION EMPRESARIAL

Red Corporativa con VLANs, HSRP, EtherChannel, STP y OSPFv2

Equipo 4 | 172.22.0.0 / 15 | 2001:25:22:: / 48
Dominio: itla.local

*HU1 Direcccionamiento | HU2 VLANs | HU3 VTP | HU4 EtherChannel+STP+HSRP | HU5 OSPFv2
Multiarea*

1. Arquitectura General y Roles de Dispositivos

La red se divide en dos zonas interconectadas por un enlace serial (s1/0 a s1/0). El dominio DNS corporativo es itla.local y se configura en todos los dispositivos.

Dispositivo	Capa	Rol	Zona
R1-Core	L3 Router	Router de borde, gateway ISP, OSPF Area 0 interno	Campus
SWM1	L3 Multilayer	Distribucion principal, HSRP Active, STP Root Primary, OSPF Area 0	Campus
SWM2	L3 Multilayer	Distribucion secundario, HSRP Standby, STP Root Secondary, OSPF Area 0	Campus
SWAC1	L2 Switch	Acceso puro L2, VLAN Docentes, VTP Client	Campus
SWAC2	L2 Switch	Acceso puro L2, VLAN Estudiantes, VTP Client	Campus
SWAC3	L2 Switch	Acceso puro L2, VLAN Gestion y Voz, VTP Client	Campus
R2	L3 Router	Router sucursal, Router-on-a-Stick, ABR OSPF Area 0 y Area 1	Sucursal
SWA13	L2 Switch	Acceso puro L2, VLANs Finanzas y Academicos, VTP Transparente	Sucursal
SW	L2 Switch	Acceso puro L2, VLAN Datacenter, VTP Transparente	Sucursal

1.1 Decision HSRP

HSRP aplica UNICAMENTE donde existe un par de dispositivos L3 actuando como gateway redundante de las mismas VLANs:

- * SWM1 y SWM2: Si aplica HSRP. Ambos son multilayer con SVIs para las VLANs del campus (10, 20, 30, 40, 50). SWM1 = Active (prioridad 110), SWM2 = Standby (prioridad 100).
- * R1-Core: NO aplica HSRP. Solo tiene enlaces L3 punto a punto hacia SWM1 y SWM2. No es gateway de ninguna VLAN de usuario.
- * R2: NO aplica HSRP. Es el unico router en la sucursal. Sin par redundante en esa zona.

1.2 Diseno de Areas OSPFv2 (2 Areas)

La division de areas que mejor se adapta a la topologia fisica es la siguiente:

Area OSPF	Tipo	Dispositivos	Justificacion
Area 0 (Backbone)	Backbone	R1-Core, SWM1, SWM2, R2 (solo interfaz s1/0)	Toda la zona del campus principal mas el enlace serial conforman el backbone. Es la zona de mayor trafico y alta disponibilidad.
Area 1 (Sucursal)	Regular	R2 (ABR)	Las redes locales de la sucursal (VLANs 60, 70, 80, 90) forman un area independiente. R2 es el unico ABR.

Esta division es mas limpia que usar 3 areas: el campus ya tiene redundancia por HSRP y EtherChannel (no necesita separarse de Area 0). R2 y sus redes locales forman una area compacta y natural.

Dispositivo	Router-ID	Rol OSPF	Areas
R1-Core	1.1.1.1	Internal Router (solo Area 0)	0

Dispositivo	Router-ID	Rol OSPF	Areas
SWM1	3.3.3.3	Internal Router (solo Area 0)	0
SWM2	4.4.4.4	Internal Router (solo Area 0)	0
R2	2.2.2.2	ABR (Area 0 via s1/0 + Area 1 via sub-interfaces)	0 y 1

2. HU1 - Plan de Direccionamiento IPv4 (VLSM)

Red base: 172.22.0.0/15. Se ordenan las redes de mayor a menor numero de hosts.

2.1 VLANs Campus (R1-Core / SWM1 / SWM2)

#	VLAN	Hosts Req.	Prefijo	Mascara	Dir. Red	1er Host	Ultimo Host	Broadcast	Hosts Usables
1	Estudiantes	1500	/21	255.255.248.0	172.22.0.0	172.22.0.1	172.22.7.254	172.22.7.255	2046
2	Docentes	164	/24	255.255.255.0	172.22.8.0	172.22.8.1	172.22.8.254	172.22.8.255	254
3	Gestion	29	/27	255.255.255.224	172.22.10.64	172.22.10.65	172.22.10.94	172.22.10.95	30
4	Voz	16	/27	255.255.255.224	172.22.10.96	172.22.10.97	172.22.10.126	172.22.10.127	30
5	Blackhole R1	10	/28	255.255.255.240	172.22.10.128	172.22.10.129	172.22.10.142	172.22.10.143	14

2.2 VLANs Sucursal (R2)

#	VLAN	Hosts Req.	Prefijo	Mascara	Dir. Red	1er Host	Ultimo Host	Broadcast	Hosts Usables
6	Academicos	80	/25	255.255.255.128	172.22.9.0	172.22.9.1	172.22.9.126	172.22.9.127	126
7	Finanzas	64	/25	255.255.255.128	172.22.9.128	172.22.9.129	172.22.9.254	172.22.9.255	126
8	Datacenter	44	/26	255.255.255.192	172.22.10.0	172.22.10.1	172.22.10.62	172.22.10.63	62
9	Blackhole R2	10	/28	255.255.255.240	172.22.10.144	172.22.10.145	172.22.10.158	172.22.10.159	14

2.3 Redes de Infraestructura P2P y Loopbacks

Enlace	Prefijo	Mascara	Dir. Red	Host A	Host B	Broadcast
R1-Core e0/0 - - SWM1 e0/0	/30	255.255.255.252	172.22.10.168	R1-Core=.169	SWM1=.170	172.22.10.171
R1-Core e0/1 - - SWM2 e0/0	/30	255.255.255.252	172.22.10.172	R1-Core=.173	SWM2=.174	172.22.10.175
R1-Core s1/0 - - R2 s1/0	/30	255.255.255.252	172.22.10.160	R1-Core=.161	R2=.162	172.22.10.163
R1-Core e0/2 - - ISP	/30	255.255.255.252	172.22.10.164	R1-Core=.165	ISP=.166	172.22.10.167
Loopback R1-Core	/32	255.255.255.255	172.22.10.176	172.22.10.176	---	---

Enlace	Prefijo	Mascara	Dir. Red	Host A	Host B	Broadcast
Loopback R2	/32	255.255.255.255	172.22.10.177	172.22.10.177	---	---
Loopback SWM1	/32	255.255.255.255	172.22.10.178	172.22.10.178	---	---
Loopback SWM2	/32	255.255.255.255	172.22.10.179	172.22.10.179	---	---

3. HU1 - Plan de Direccionamiento IPv6

Bloque: 2001:25:22::/48. Prefijos /64 asignados en orden descendente de tamaño IPv4. Los hosts obtienen su IP por SLAAC.

Orden	VLAN / Enlace	Red IPv4	Prefijo IPv6 /64	Gateway A	Gateway B
1	Estudiantes	172.22.0.0/21	2001:25:22:1::/64	SWM1=::2 / VIP HSRP=::1	SWM2=::3
2	Docentes	172.22.8.0/24	2001:25:22:2::/64	SWM1=::2 / VIP HSRP=::1	SWM2=::3
3	Academicos	172.22.9.0/25	2001:25:22:3::/64	R2 e0/0.70=::1	---
4	Finanzas	172.22.9.128/25	2001:25:22:4::/64	R2 e0/0.60=::1	---
5	Datacenter	172.22.10.0/26	2001:25:22:5::/64	R2 e0/1.80=::1	---
6	Gestion	172.22.10.64/27	2001:25:22:6::/64	SWM1=::2 / VIP HSRP=::1	SWM2=::3
7	Voz	172.22.10.96/27	2001:25:22:7::/64	SWM1=::2 / VIP HSRP=::1	SWM2=::3
8	Blackhole R1	172.22.10.128/28	2001:25:22:8::/64	SWM1=::2 / VIP HSRP=::1	SWM2=::3
9	Blackhole R2	172.22.10.144/28	2001:25:22:9::/64	R2=::1	---
10	P2P R1-R2 serial	172.22.10.160/30	2001:25:22:A::/64	R1-Core s1/0=::1	R2 s1/0=::2
11	R1-Core a ISP	172.22.10.164/30	2001:25:22:B::/64	R1-Core e0/2=::1	ISP=::2
12	R1-Core a SWM1	172.22.10.168/30	2001:25:22:C::/64	R1-Core e0/0=::1	SWM1=::2
13	R1-Core a SWM2	172.22.10.172/30	2001:25:22:D::/64	R1-Core e0/1=::1	SWM2=::2

4. HU2 - Segmentacion VLAN

ID	Nombre	Zona	Hosts	GW IPv4 (HSRP VIP)	GW IPv6	Acceso
10	Estudiantes	Campus	1500	172.22.0.1	2001:25:22:1::1	SWAC2 e2/0
20	Docentes	Campus	164	172.22.8.1	2001:25:22:2::1	SWAC1 e1/0
30	Gestion	Campus	29	172.22.10.65	2001:25:22:6::1	SWAC3 e3/0
40	Voz	Campus	16	172.22.10.97	2001:25:22:7::1	SWAC3 e3/0 (voice)
50	Blackhole R1	Campus	10	172.22.10.129	2001:25:22:8::1	Sin hosts
60	Finanzas	Sucursal	64	172.22.9.129	2001:25:22:4::1	SWA13 e2/0
70	Academicos	Sucursal	80	172.22.9.1	2001:25:22:3::1	SWA13 e1/0
80	Datacenter	Sucursal	44	172.22.10.1	2001:25:22:5::1	SW e0/1
90	Blackhole R2	Sucursal	10	172.22.10.145	2001:25:22:9::1	Sin hosts
99	Native Mgmt	Global	---	---	---	VLAN nativa trunks

5. HU3 - VTP

Dispositivo	Dominio	Modo	Version	Password	Razon
SWM1	EMPRESA	Server	2	cisco123	Crea y propaga VLANs 10,20,30,40,50,99 al campus
SWM2	EMPRESA	Server	2	cisco123	Servidor VTP redundante junto con SWM1
SWAC1	EMPRESA	Client	2	cisco123	Recibe VLANs, no puede crearlas ni eliminarlas
SWAC2	EMPRESA	Client	2	cisco123	Recibe VLANs, no puede crearlas ni eliminarlas
SWAC3	EMPRESA	Client	2	cisco123	Recibe VLANs, no puede crearlas ni eliminarlas
SWA13	R2BRANCH	Transparent	2	---	VLANs 60,70 configuradas localmente. No hay trunk continuo desde campus.
SW	R2BRANCH	Transparent	2	---	VLAN 80 configurada localmente

6. HU4 - EtherChannel, Rapid PVST+ y HSRP

6.1 EtherChannel LACP

Parametro	Valor
Protocolo	LACP IEEE 802.3ad
Port-Channel	Po1 en SWM1 y SWM2
Modo	SWM1: active / SWM2: active
Interfaces fisicas	e0/1, e0/2, e0/3 en ambos switches
Tipo de trunk	dot1Q
VLANs permitidas	10, 20, 30, 40, 50, 99
VLAN nativa	99

6.2 Rapid PVST+

VLAN	Root Primary	Prioridad	Root Secondary	Prioridad
10 Estudiantes	SWM1	4096	SWM2	8192
20 Docentes	SWM1	4096	SWM2	8192
30 Gestion	SWM1	4096	SWM2	8192
40 Voz	SWM1	4096	SWM2	8192
50 Blackhole R1	SWM1	4096	SWM2	8192

Puertos de acceso en SWAC1/2/3: PortFast + BPDU Guard activados en todos los puertos de acceso.

6.3 HSRP v2 (solo en SWM1 y SWM2)

Grupo	VLAN	VIP Gateway	SWM1 IP (Active-110)	SWM2 IP (Standby-100)	Preempt	Timers
10	10 Estudiantes	172.22.0.1	172.22.0.2	172.22.0.3	Si	hello 3s / hold 10s
20	20 Docentes	172.22.8.1	172.22.8.2	172.22.8.3	Si	hello 3s / hold 10s
30	30 Gestion	172.22.10.65	172.22.10.66	172.22.10.67	Si	hello 3s / hold 10s
40	40 Voz	172.22.10.97	172.22.10.98	172.22.10.99	Si	hello 3s / hold 10s
50	50 Blackhole	172.22.10.129	172.22.10.130	172.22.10.131	Si	hello 3s / hold 10s

7. HU5 - OSPFv2 Multiarea (2 Areas)

7.1 Resumen de Areas

Area	Tipo	Redes	Dispositivos Participantes
0 - Backbone	Backbone	172.22.10.160/30 (serial P2P), 172.22.10.168/30 (R1-SWM1), 172.22.10.172/30 (R1-SWM2), VLANs 10/20/30/40/50, Loopbacks R1/SWM1/SWM2	R1-Core, SWM1, SWM2 (internos) + R2 (solo s1/0 en Area 0)
1 - Sucursal	Regular	VLANs 60/70/80/90 (redes locales R2), Loopback R2	R2 como ABR unico. SWA13 y SW no participan en OSPF (son L2).

7.2 Ruta Predeterminada

R1-Core tiene una ruta estatica hacia el ISP (0.0.0.0/0 via 172.22.10.166). Se redistribuye con default-information originate, generando un LSA tipo 5 que llega a SWM1, SWM2 y R2.

8. Configuracion IP de las Computadoras (End Hosts)

Las siguientes IPs se asignan manualmente en cada PC dentro de PNETLab. El DNS se configura apuntando al servidor del Datacenter o al gateway.

PC / Host	VLAN	IP Asignada	Mascara	Gateway (HSRP VIP o router)	DNS	IPv6 (SLAAC)
DOC1	20 - Docentes	172.22.8.10	255.255.255.0	172.22.8.1	172.22.10.1	2001:25:22:2::/64 (auto)
EST1	10 - Estudiantes	172.22.0.10	255.255.248.0	172.22.0.1	172.22.10.1	2001:25:22:1::/64 (auto)
GES1	30 - Gestion	172.22.10.70	255.255.255.224	172.22.10.65	172.22.10.1	2001:25:22:6::/64 (auto)
ACA1	70 - Academicos	172.22.9.10	255.255.255.128	172.22.9.1	172.22.10.1	2001:25:22:3::/64 (auto)
FIN1	60 - Finanzas	172.22.9.140	255.255.255.128	172.22.9.129	172.22.10.1	2001:25:22:4::/64 (auto)
SERVICIOS	80 - Datacenter	172.22.10.10	255.255.255.192	172.22.10.1	172.22.10.10	2001:25:22:5::/64 (auto)

Nota: SERVICIOS actua como servidor (Datacenter). Su propia IP se usa como DNS para las demas VLANs. Los hosts de las VLANs del campus usan la VIP de HSRP como gateway (nunca la IP real de SWM1 o SWM2 directamente).

Direccionamientos dispositivos

PC	IP	Máscara	Gateway	VLAN
DOC1	172.22.8.10	255.255.255.0	172.22.8.1	20 - Docentes
EST1	172.22.0.10	255.255.248.0	172.22.0.1	10 - Estudiantes
GES1	172.22.10.70	255.255.255.224	172.22.10.65	30 - Gestión
ACA1	172.22.9.10	255.255.255.128	172.22.9.1	70 - Académicos
FIN1	172.22.9.140	255.255.255.128	172.22.9.129	60 - Finanzas
SERVICIOS-DOCKER	172.22.10.10	255.255.255.192	172.22.10.1	80 - Datacenter

9. Scripts de Configuración por Dispositivo

Scripts listos para pegar directamente en la consola de PNETLab. Sin caracteres especiales ni Unicode. Los comentarios inician con ! (estandar Cisco IOS). Las líneas con sangría pertenecen a un subcomando.

9.1 R1-Core

```
enable
configure terminal
hostname R1-Core
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
*   Solo acceso autorizado - EQUIPO 4   *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
ipv6 unicast-routing
interface e0/0
no shutdown
ip address 172.22.10.169 255.255.255.252
ipv6 address 2001:25:22:C::1/64
ipv6 enable
duplex full
description Enlace-L3-a-SWM1
interface e0/1
no shutdown
ip address 172.22.10.173 255.255.255.252
ipv6 address 2001:25:22:D::1/64
ipv6 enable
duplex full
description Enlace-L3-a-SWM2
interface e0/2
no shutdown
ip address 172.22.10.165 255.255.255.252
ipv6 address 2001:25:22:B::1/64
```

```

    ipv6 enable
    description Enlace-a-ISP
interface s1/0
    no shutdown
    ip address 172.22.10.161 255.255.255.252
    ipv6 address 2001:25:22:A::1/64
    ipv6 enable
    clock rate 64000
    description Serial-WAN-a-R2
interface Loopback0
    ip address 172.22.10.176 255.255.255.255
    description Loopback-R1Core
ip route 0.0.0.0 0.0.0.0 172.22.10.166
router ospf 1
    router-id 1.1.1.1
    network 172.22.10.160 0.0.0.3 area 0
    network 172.22.10.168 0.0.0.3 area 0
    network 172.22.10.172 0.0.0.3 area 0
    network 172.22.10.176 0.0.0.0 area 0
    default-information originate
    passive-interface e0/2
end
write memory

```

9.2 R2 - Router Sucursal

```

enable
configure terminal
hostname R2
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
*      Solo acceso autorizado - EQUIPO 4      *
*****
#
line console 0
    password cisco123
    login
    logging synchronous
line vty 0 4
    password cisco123
    login
    transport input telnet ssh
ipv6 unicast-routing
interface s1/0
    no shutdown
    ip address 172.22.10.162 255.255.255.252
    ipv6 address 2001:25:22:A::2/64
    ipv6 enable
    description Serial-WAN-a-R1Core
interface e0/0
    no shutdown
    description Trunk-a-SWA13
interface e0/0.70
    encapsulation dot1Q 70
    ip address 172.22.9.1 255.255.255.128
    ipv6 address 2001:25:22:3::1/64
    description VLAN-70-Academicos
interface e0/0.60
    encapsulation dot1Q 60
    ip address 172.22.9.129 255.255.255.128
    ipv6 address 2001:25:22:4::1/64
    description VLAN-60-Finanzas
interface e0/0.90
    encapsulation dot1Q 90
    ip address 172.22.10.145 255.255.255.240
    ipv6 address 2001:25:22:9::1/64
    description VLAN-90-Blackhole-R2
interface e0/1
    no shutdown

```

```

    description Trunk-a-SW-Datacenter
interface e0/1.80
    encapsulation dot1Q 80
    ip address 172.22.10.1 255.255.255.192
    ipv6 address 2001:25:22:5::1/64
    description VLAN-80-Datacenter
interface Loopback0
    ip address 172.22.10.177 255.255.255.255
    description Loopback-R2
router ospf 1
    router-id 2.2.2.2
    network 172.22.10.160 0.0.0.3 area 0
    network 172.22.10.177 0.0.0.0 area 0
    network 172.22.9.0 0.0.0.127 area 1
    network 172.22.9.128 0.0.0.127 area 1
    network 172.22.10.0 0.0.0.63 area 1
    network 172.22.10.144 0.0.0.15 area 1
    passive-interface e0/0.70
    passive-interface e0/0.60
    passive-interface e0/0.90
    passive-interface e0/1.80
end
write memory

```

9.3 SWM1 - Switch Multilayer Distribución Principal

```

enable
configure terminal
hostname SWM1
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
*   Solo acceso autorizado - EQUIPO 4   *
*****
#
line console 0
    password cisco123
    login
    logging synchronous
line vty 0 4
    password cisco123
    login
    transport input telnet ssh
ip routing
ipv6 unicast-routing
vtp domain EMPRESA
vtp mode server
vtp version 2
vtp password cisco123
vlan 10
    name Estudiantes
vlan 20
    name Docentes
vlan 30
    name Gestion
vlan 40
    name Voz
vlan 50
    name Blackhole_R1
vlan 99
    name Native_Mgmt
spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40,50 priority 4096
interface range e0/1-3
    switchport trunk encapsulation dot1q
    switchport mode trunk
    switchport trunk native vlan 99
    switchport trunk allowed vlan 10,20,30,40,50,99

```

```
channel-group 1 mode active
channel-protocol lacp
no shutdown
interface Port-channel1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,30,40,50,99
description EtherChannel-LACP-a-SWM2
no shutdown
interface e0/0
no switchport
ip address 172.22.10.170 255.255.255.252
ipv6 address 2001:25:22:C::2/64
ipv6 enable
duplex full
description Enlace-L3-a-R1Core
no shutdown
interface e1/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 20,99
description Trunk-a-SWAC1
no shutdown
interface e1/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,99
description Trunk-a-SWAC2
no shutdown
interface e1/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 30,40,99
description Trunk-a-SWAC3
no shutdown
interface Vlan10
ip address 172.22.0.2 255.255.248.0
ipv6 address 2001:25:22:1::2/64
ipv6 enable
standby version 2
standby 10 ip 172.22.0.1
standby 10 priority 110
standby 10 preempt
standby 10 timers 3 10
description SVI-VLAN10-Estudiantes
no shutdown
interface Vlan20
ip address 172.22.8.2 255.255.255.0
ipv6 address 2001:25:22:2::2/64
ipv6 enable
standby version 2
standby 20 ip 172.22.8.1
standby 20 priority 110
standby 20 preempt
standby 20 timers 3 10
description SVI-VLAN20-Docentes
no shutdown
interface Vlan30
ip address 172.22.10.66 255.255.255.224
ipv6 address 2001:25:22:6::2/64
ipv6 enable
standby version 2
standby 30 ip 172.22.10.65
standby 30 priority 110
standby 30 preempt
standby 30 timers 3 10
description SVI-VLAN30-Gestion
no shutdown
interface Vlan40
ip address 172.22.10.98 255.255.255.224
ipv6 address 2001:25:22:7::2/64
ipv6 enable
standby version 2
standby 40 ip 172.22.10.97
```

```

standby 40 priority 110
standby 40 preempt
standby 40 timers 3 10
description SVI-VLAN40-Voz
no shutdown
interface Vlan50
ip address 172.22.10.130 255.255.255.240
ipv6 address 2001:25:22:8::2/64
ipv6 enable
standby version 2
standby 50 ip 172.22.10.129
standby 50 priority 110
standby 50 preempt
standby 50 timers 3 10
description SVI-VLAN50-Blackhole-R1
no shutdown
interface Loopback0
ip address 172.22.10.178 255.255.255.255
description Loopback-SWM1
router ospf 1
router-id 3.3.3.3
network 172.22.0.0 0.0.7.255 area 0
network 172.22.8.0 0.0.0.255 area 0
network 172.22.10.64 0.0.0.31 area 0
network 172.22.10.96 0.0.0.31 area 0
network 172.22.10.128 0.0.0.15 area 0
network 172.22.10.168 0.0.0.3 area 0
network 172.22.10.178 0.0.0.0 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan50
end
write memory

```

9.4 SWM2 - Switch Multilayer Distribución Secundario

```

enable
configure terminal
hostname SWM2
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
*   Solo acceso autorizado - EQUIPO 4   *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
ip routing
ipv6 unicast-routing
vtp domain EMPRESA
vtp mode server
vtp version 2
vtp password cisco123
vlan 10
name Estudiantes
vlan 20
name Docentes
vlan 30
name Gestion
vlan 40
name Voz
vlan 50
name Blackhole_R1

```

```
vlan 99
  name Native_Mgmt
  spanning-tree mode rapid-pvst
  spanning-tree vlan 10,20,30,40,50 priority 8192
interface range e0/1-3
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,50,99
  channel-group 1 mode active
  channel-protocol lacp
  no shutdown
interface Port-channel1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,50,99
  description EtherChannel-LACP-a-SWM1
  no shutdown
interface e0/0
  no switchport
  ip address 172.22.10.174 255.255.255.252
  ipv6 address 2001:25:22:D::2/64
  ipv6 enable
  duplex full
  description Enlace-L3-a-R1Core
  no shutdown
interface e1/0
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 20,99
  description Trunk-a-SWAC1
  no shutdown
interface e1/1
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,99
  description Trunk-a-SWAC2
  no shutdown
interface e1/2
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 30,40,99
  description Trunk-a-SWAC3
  no shutdown
interface Vlan10
  ip address 172.22.0.3 255.255.248.0
  ipv6 address 2001:25:22:1::3/64
  ipv6 enable
  standby version 2
  standby 10 ip 172.22.0.1
  standby 10 priority 100
  standby 10 preempt
  standby 10 timers 3 10
  description SVI-VLAN10-Estudiantes
  no shutdown
interface Vlan20
  ip address 172.22.8.3 255.255.255.0
  ipv6 address 2001:25:22:2::3/64
  ipv6 enable
  standby version 2
  standby 20 ip 172.22.8.1
  standby 20 priority 100
  standby 20 preempt
  standby 20 timers 3 10
  description SVI-VLAN20-Docentes
  no shutdown
interface Vlan30
  ip address 172.22.10.67 255.255.255.224
  ipv6 address 2001:25:22:6::3/64
  ipv6 enable
  standby version 2
  standby 30 ip 172.22.10.65
  standby 30 priority 100
  standby 30 preempt
```

```

standby 30 timers 3 10
description SVI-VLAN30-Gestion
no shutdown
interface Vlan40
ip address 172.22.10.99 255.255.255.224
ipv6 address 2001:25:22:7::3/64
ipv6 enable
standby version 2
standby 40 ip 172.22.10.97
standby 40 priority 100
standby 40 preempt
standby 40 timers 3 10
description SVI-VLAN40-Voz
no shutdown
interface Vlan50
ip address 172.22.10.131 255.255.255.240
ipv6 address 2001:25:22:8::3/64
ipv6 enable
standby version 2
standby 50 ip 172.22.10.129
standby 50 priority 100
standby 50 preempt
standby 50 timers 3 10
description SVI-VLAN50-Blackhole-R1
no shutdown
interface Loopback0
ip address 172.22.10.179 255.255.255.255
description Loopback-SWM2
router ospf 1
router-id 4.4.4.4
network 172.22.0.0 0.0.7.255 area 0
network 172.22.8.0 0.0.0.255 area 0
network 172.22.10.64 0.0.0.31 area 0
network 172.22.10.96 0.0.0.31 area 0
network 172.22.10.128 0.0.0.15 area 0
network 172.22.10.172 0.0.0.3 area 0
network 172.22.10.179 0.0.0.0 area 0
passive-interface Vlan10
passive-interface Vlan20
passive-interface Vlan30
passive-interface Vlan40
passive-interface Vlan50
end
write memory

```

9.5 SWAC1 - Switch Acceso L2 (Docentes)

```

enable
configure terminal
hostname SWAC1
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
*      Solo acceso autorizado - EQUIPO 4      *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
vtp domain EMPRESA
vtp mode client
vtp version 2
vtp password cisco123
spanning-tree mode rapid-pvst
interface e0/0

```

```

switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 20,99
description Trunk-a-SWM1
no shutdown
interface e0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 20,99
description Trunk-a-SWM2
no shutdown
interface e1/0
switchport mode access
switchport access vlan 20
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-DOC1-VLAN20
no shutdown
end
write memory

```

9.6 SWAC2 - Switch Acceso L2 (Estudiantes)

```

enable
configure terminal
hostname SWAC2
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
*   Solo acceso autorizado - EQUIPO 4   *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
vtp domain EMPRESA
vtp mode client
vtp version 2
vtp password cisco123
spanning-tree mode rapid-pvst
interface e0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,99
description Trunk-a-SWM1
no shutdown
interface e0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,99
description Trunk-a-SWM2
no shutdown
interface e2/0
switchport mode access
switchport access vlan 10

```



```
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-EST1-VLAN10
no shutdown
end
write memory
```

9.7 SWAC3 - Switch Acceso L2 (Gestion y Voz)

```
enable
configure terminal
hostname SWAC3
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
*   Solo acceso autorizado - EQUIPO 4   *
*****
#
line console 0
password cisco123
login
logging synchronous
line vty 0 4
password cisco123
login
transport input telnet ssh
vtp domain EMPRESA
vtp mode client
vtp version 2
vtp password cisco123
spanning-tree mode rapid-pvst
interface e0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 30,40,99
description Trunk-a-SWM1
no shutdown
interface e0/1
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 30,40,99
description Trunk-a-SWM2
no shutdown
interface e3/0
switchport mode access
switchport access vlan 30
switchport voice vlan 40
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-GES1-Data30-Voice40
no shutdown
end
write memory
```

9.8 SWA13 - Switch Acceso Sucursal (Finanzas y Academicos)

```
enable
configure terminal
hostname SWA13
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
```

```

*****
*   Solo acceso autorizado - EQUIPO 4   *
*****
#
line console 0
  password cisco123
  login
  logging synchronous
line vty 0 4
  password cisco123
  login
  transport input telnet ssh
vtp domain R2BRANCH
vtp mode transparent
vtp version 2
vlan 60
  name Finanzas
vlan 70
  name Academicos
vlan 99
  name Native_Mgmt
spanning-tree mode rapid-pvst
interface e0/0
  switchport trunk encapsulation dot1q
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 60,70,99
  description Trunk-a-R2-e0/0
  no shutdown
interface e1/0
  switchport mode access
  switchport access vlan 70
  spanning-tree portfast
  spanning-tree bpduguard enable
  description Acceso-ACA1-VLAN70
  no shutdown
interface e2/0
  switchport mode access
  switchport access vlan 60
  spanning-tree portfast
  spanning-tree bpduguard enable
  description Acceso-FIN1-VLAN60
  no shutdown
end
write memory

```

9.9 SW - Switch Acceso Sucursal (Datacenter)

```

enable
configure terminal
hostname SW
no ip domain-lookup
ip domain-name itla.local
service password-encryption
enable secret cisco123
banner motd #
*****
*   Solo acceso autorizado - EQUIPO 4   *
*****
#
line console 0
  password cisco123
  login
  logging synchronous
line vty 0 4
  password cisco123
  login
  transport input telnet ssh
vtp domain R2BRANCH
vtp mode transparent
vtp version 2
vlan 80
  name Datacenter
vlan 99

```

```
name Native_Mgmt
spanning-tree mode rapid-pvst
interface e0/0
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 80,99
description Trunk-a-R2-e0/1
no shutdown
interface e0/1
switchport mode access
switchport access vlan 80
spanning-tree portfast
spanning-tree bpduguard enable
description Acceso-SERVICIOS-VLAN80
no shutdown
end
write memory
```

10. Comandos de Verificacion

VLANs y VTP

```
show vlan brief
show vtp status
show interfaces trunk
show etherchannel summary
```

STP y HSRP

```
show spanning-tree vlan 10
show spanning-tree summary
show standby brief
show standby vlan 10
```

OSPFv2

```
show ip ospf neighbor
show ip route ospf
show ip ospf database
show ip ospf border-routers
show ip route 0.0.0.0
```

Conectividad

```
! Desde DOC1 (172.22.8.10) hacia gateway HSRP:
ping 172.22.8.1
! Hacia EST1 (inter-VLAN via SWM):
ping 172.22.0.10
! Hacia sucursal (via OSPF Area 0 a Area 1):
ping 172.22.9.10
traceroute 172.22.9.1
! Hacia Internet:
ping 8.8.8.8
```

11. Resumen Final HU1 a HU5

HU	Tecnologia	Detalles	Estado
HU1 IPv4	VLSM 172.22.0.0/15	13 subredes, mayor a menor, 9 VLANs + 4 infraestructura	Completo
HU1 IPv6	Prefijos /64 2001:25:22::/48	13 prefijos asignados en orden de tamaño IPv4, SLAAC	Completo
HU2	VLANs dot1Q	10 VLANs (10-99), mapa de puertos por dispositivo	Completo
HU3	VTP v2	Dominio EMPRESA (Server/Client) + R2BRANCH (Transparente)	Completo
HU4 L2	EtherChannel LACP + Rapid PVST+	Po1 3 enlaces, SWM1 Root Primary, PortFast+BPDUGuard	Completo
HU4 L3	HSRP v2	5 grupos en SWM1/SWM2, failover menor a 10 segundos	Completo
HU5	OSPFv2 2 Areas	Area 0=Campus, Area 1=Sucursal, R2 es ABR, default-information originate	Completo
HU6	ACLs	Pendiente Sprint 3	Pendiente